

# Research in Computer Science

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## Research in Computer Science

Research in CS is very diverse

Which means that goals, methodologies, hypothesis, assumptions,... may be very different

# Types of Research

Rough classification (many research works combine many of them):



- **Formal or Theoretical:** Proof facts
- **Build or Design:** Building an artifact
- **Process:** Understand the process followed by users to accomplish tasks in CS
- **Model:** Defining abstract models for real systems

In all of them there may be an **experimental** part.

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- Often times is **basic research**
- Because of its precision, sometimes it provides the sharpest results

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- **Understanding quantum computing**

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- **Proof of concept or prototype** (demonstrate that it is possible)
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- Achieving **quantum supremacy**

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- Essentially to achieve Maximum scientific rigor possible:
  - **Research Methodology must be careful with:**
    - Hypothesis: falsifiable, testable
    - Experiments: replicable, objective (procedures, instruments, measurements, samples,...)
    - **Results Interpretation:** logically consistent

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- **Tools for quantum programming**

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## Example:

- Balancing Bike Sharing Systems

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from the webpage of Gloria Mark at UCI

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Process Research, Observational, Quantitative

# Example 1 (boolean SATisfiability)

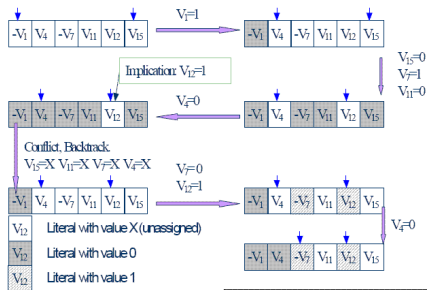


Figure 1: BCP using two watched literals

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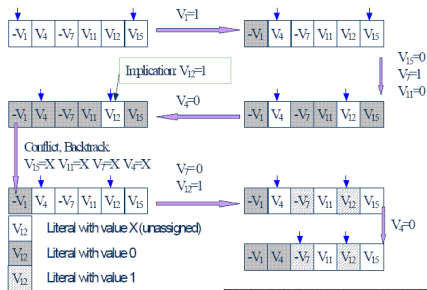


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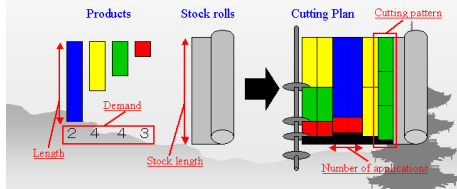
**Invent a new data structure that makes SAT solvers more efficient**

- Build (it only affects one module of the algorithm)
- Quantitative (measures performance)
- Experimental (using benchmarks)

# Example 2 (cutting stock)

## One Dimensional Cutting Stock Problem (1D-CSP)

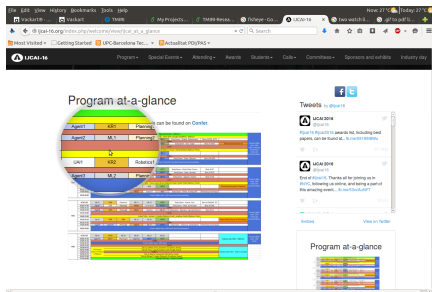
- 1D-CSP asks to specify how stock rolls can be cut into products while satisfying all demands.
- A cutting plan (solution) includes two factors:
  - A set of cutting patterns
  - Number of applications of cutting patterns.



## Model and solve with Constraint Programming for the first time

- Model
- Quantitative
- Experimental (using benchmarks)

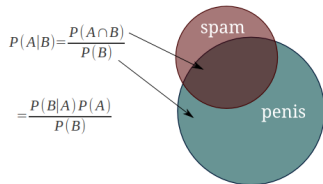
# Example 3 (file navigation in software development)



**Propose a fisheye graphical interface to navigate on the files of a project**

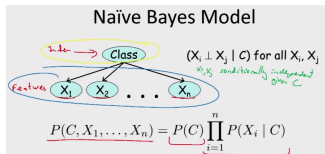
- Build and Process
- Quantitative
- Experimental (measure concentration, errors,...)

## Example 4 (spam filters)

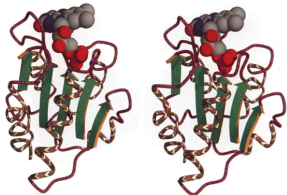
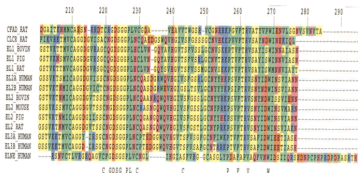


**Propose a new prediction model to identify spam**

- Model
- Quantitative
- experimental



# Example 5 (protein alignment)



## Propose a new model for protein alignment

- Model
- Maybe qualitative if it is supported by biological narrative
- Maybe observational if it is supported by consistency with respect to observed real proteins